

空间曲线 C , 切点 P_0 , 切向量为
 $(\varphi'(t_0), \psi'(t_0), w'(t_0))$

曲线 C 在点 $P_0(\varphi(t_0), \psi(t_0), w(t_0))$ 的切线方程为

$$\frac{x - \varphi(t_0)}{\varphi'(t_0)} = \frac{y - \psi(t_0)}{\psi'(t_0)} = \frac{z - w(t_0)}{w'(t_0)}$$

曲线 C 在点 P_0 的法平面方程

$$\varphi'(t_0)(x - \varphi(t_0)) + \psi'(t_0)(y - \psi(t_0)) + w'(t_0)(z - w(t_0)) = 0$$

曲线一般式方程

$$C = \begin{cases} F(x, y, z) = 0 \\ H(x, y, z) = 0 \end{cases} \quad \text{设 } C = \vec{r}(t) = (\varphi(t), \psi(t), w(t))$$

$$\begin{cases} F(\varphi(t), \psi(t), w(t)) \equiv 0 \\ H(\varphi(t), \psi(t), w(t)) \equiv 0 \end{cases} \Rightarrow \begin{cases} \nabla F \cdot \vec{r}'(t) = 0 \\ \nabla H \cdot \vec{r}'(t) = 0 \end{cases}$$

$$\Rightarrow \vec{r}'(t) \parallel \nabla F \times \nabla H$$

法平面方程

$$\begin{vmatrix} x - x_0 & y - y_0 & z - z_0 \\ F'_x(P_0) & F'_y(P_0) & F'_z(P_0) \\ H'_x(P_0) & H'_y(P_0) & H'_z(P_0) \end{vmatrix} = 0$$

例 $\begin{cases} x^2+y^2+z^2=6 \\ x+y+z=4 \end{cases}$ 在 $(1,1,2)$ 处切线和法平面

$$F = x^2+y^2+z^2=6 \quad H = x+y+z=4$$

$$\nabla F|_{(1,1,2)} = (2x, 2y, 2z) = (2, 2, 4) \quad \nabla H|_{(1,1,2)} = (1, 1, 1)$$

$$\text{切方向} = \nabla F \times \nabla H = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 2 & 4 \\ 1 & 1 & 1 \end{vmatrix} = (-1, 1, 0)$$

$$\Rightarrow \text{切线} \quad \frac{x-1}{-1} = \frac{y-1}{1} = \frac{z-2}{0}$$

$$\text{法平面} \quad x-y=0$$

曲面 S 的一般式方程 $F(x, y, z) = 0$, F 连续可微, 且 F'_x, F'_y, F'_z 不全为 0

过点 P_0 取一条光滑曲线 $F(\varphi(t), \psi(t), w(t)) \equiv 0$

$$\Rightarrow r' = (\varphi'(t), \psi'(t), w'(t))$$

$$\nabla F(P_0) \cdot \vec{r}'(t_0) = 0$$

$$n = (F'_x, F'_y, F'_z)$$

向量 n 是空间曲面 S 在点 P_0 的法向量

$$\text{切平面方程} \quad F'_x(P_0)(x-x_0) + F'_y(P_0)(y-y_0) + F'_z(P_0)(z-z_0) = 0$$

$$\text{法线方程} \quad \frac{x-x_0}{F'_x(P_0)} = \frac{y-y_0}{F'_y(P_0)} = \frac{z-z_0}{F'_z(P_0)}$$

曲面 S 的参数方程

$$S: \vec{r}(u,v) = (x(u,v), y(u,v), z(u,v))$$

$$F(x(u,v), y(u,v), z(u,v)) \equiv 0$$

$$\begin{cases} F'_x x'_u + F'_y y'_u + F'_z z'_u = 0 \\ F'_x x'_v + F'_y y'_v + F'_z z'_v = 0 \end{cases}$$

$$\text{记 } \vec{r}(u,v) = (x(u,v), y(u,v), z(u,v)),$$

$$\text{法向量 } \vec{r}'_u \times \vec{r}'_v = (A, B, C) = \left(\frac{Dy, z}{D(u,v)}, \frac{Dz, x}{D(u,v)}, \frac{D(x, y)}{D(u,v)} \right)$$

$$\text{法线方程 } \frac{x-x_0}{A(u_0, v_0)} = \frac{y-y_0}{B(u_0, v_0)} = \frac{z-z_0}{C(u_0, v_0)}$$

例: 求曲面 $x^2 + y^2 = 4z$ 的切平面, 使它通过曲线 $x=t^2, y=t, z=3t-3$ 在 $t=1$ 处的切线

$$\text{切向量 } l(2, 1, 3)$$

$$\text{法向量 } n(2x, 2y, -4) \quad \text{切平面 } 2x_0(x-x_0) + 2y_0(y-y_0) - 4(z-z_0) = 0$$

$$\text{而 } x_0^2 + y_0^2 = 4z_0 \Rightarrow x_0 x + y_0 y - 2z - 2z_0 = 0$$

$$\text{过点 } (1, 1, 0) \Rightarrow x_0 + y_0 - 2z_0 = 0$$

$$l \cdot n = 0 \Rightarrow 4x_0 + 2y_0 - 12 = 0$$

$$\Rightarrow x + y - z - 2 = 0 \quad \text{或} \quad 6x + 3y - 5z - 9 = 0$$